Investigating processes in solid catalysts on different time scales with the periodic transient kinetic method

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The understanding of catalysts under dynamic reaction conditions is necessary for future development of applications [1]. Recently, Borgschulte et al. came up with the term 'catalytic impedance spectroscopy' [2]. The described methodology features some concepts which are already established experimentally and published by our group more than 3 years ago under the terminology periodic transient kinetic (PTK) method. In this contribution, the features of the PTK method are illustrated.

The PTK method can only unfold its full potential, if the technical equipment is matched with the intended experimental outcome. Therefore, we developed a setup for fast and stable periodic switching between two gas streams allowing for period durations as low as 6 s. The setup consists of a reactor with an aluminum jacket assuring kinetically controlled measurements under isothermal conditions. Furthermore, the reactor can be operated under elevated pressure, while pressure shocks during switching of gas streams are avoided. Additionally, it allows for quantitative analysis of the transient mass balance by internal and external standards with a temporal resolution of ca. 0.5 s [3]. The method was successfully applied for investigation of fast reactions in micro-kinetic analysis, where the quantitative and transient measurement of individual gas phase components has been proven [4]. Moreover, the technique provides insights into the reaction mechanism by providing the experimental data basis for fitting of micro-kinetic models (see fig. 1).

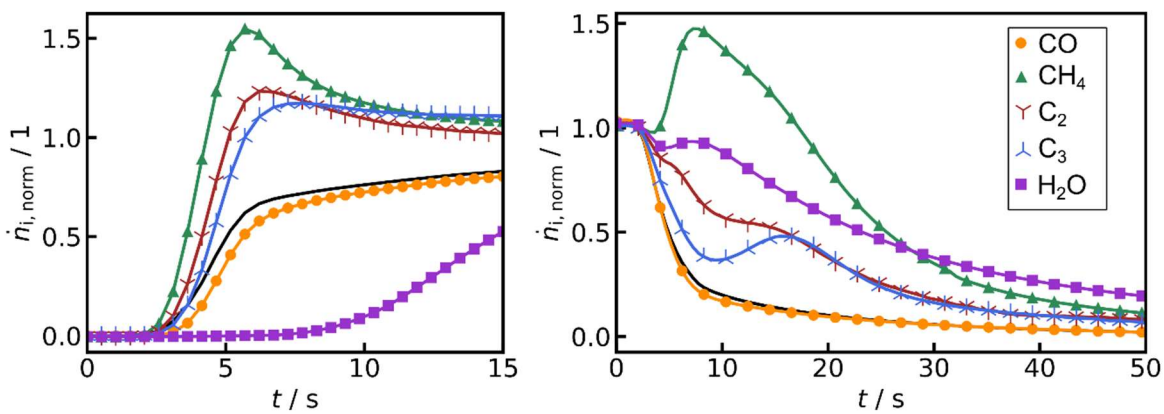

Figure 1: Measured Build-up phase (left) and back-transient phase (right) during CO hydrogenation on Co/TiO_2 (adapted from [5]).

The PTK method can also be used for investigating slow processes, e.g. for deactivation of solid catalysts [6]. In this case, the steep responses after the switching are still present, but the focus of evaluation are the pseudo steady-states at the end of each half-period (see fig.2a, region 100-120 s). This point can be used to track the activity of the catalyst over the reaction time of the periodic experiment. The activity of the catalyst is measured for the periodic operation under PTK- and steady-state operation. The results are depicted in fig.2b), where it can be seen that the activation loss differs for different operation modes. From this information, it can be deduced whether the deactivation is mainly thermally or chemically induced. For the shown CO methanation, the deactivation can be classified as mainly reversible chemically induced deactivation as it is known from literature [7].

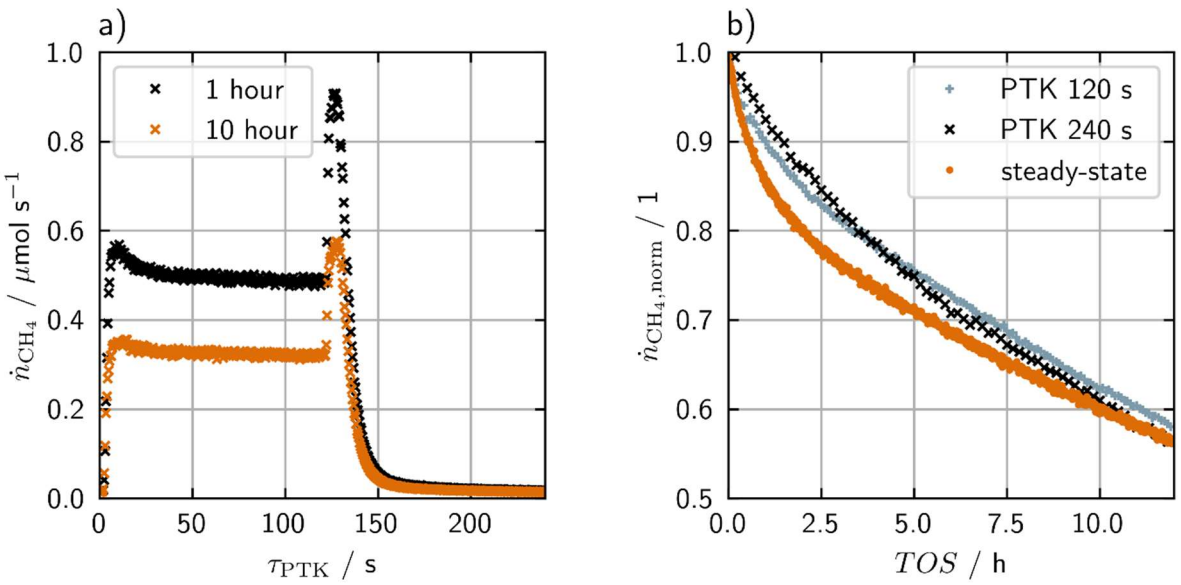

Figure 2: a) CH_4 formation rate in a single period of CO-methanation on Ni/Al_2O_3 catalyst after 1 hour and 10 hours of operation, b) Normalized CH_4 formation rate over the time on stream (TOS) for steady-state and PTK experiments with different period durations [6].

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